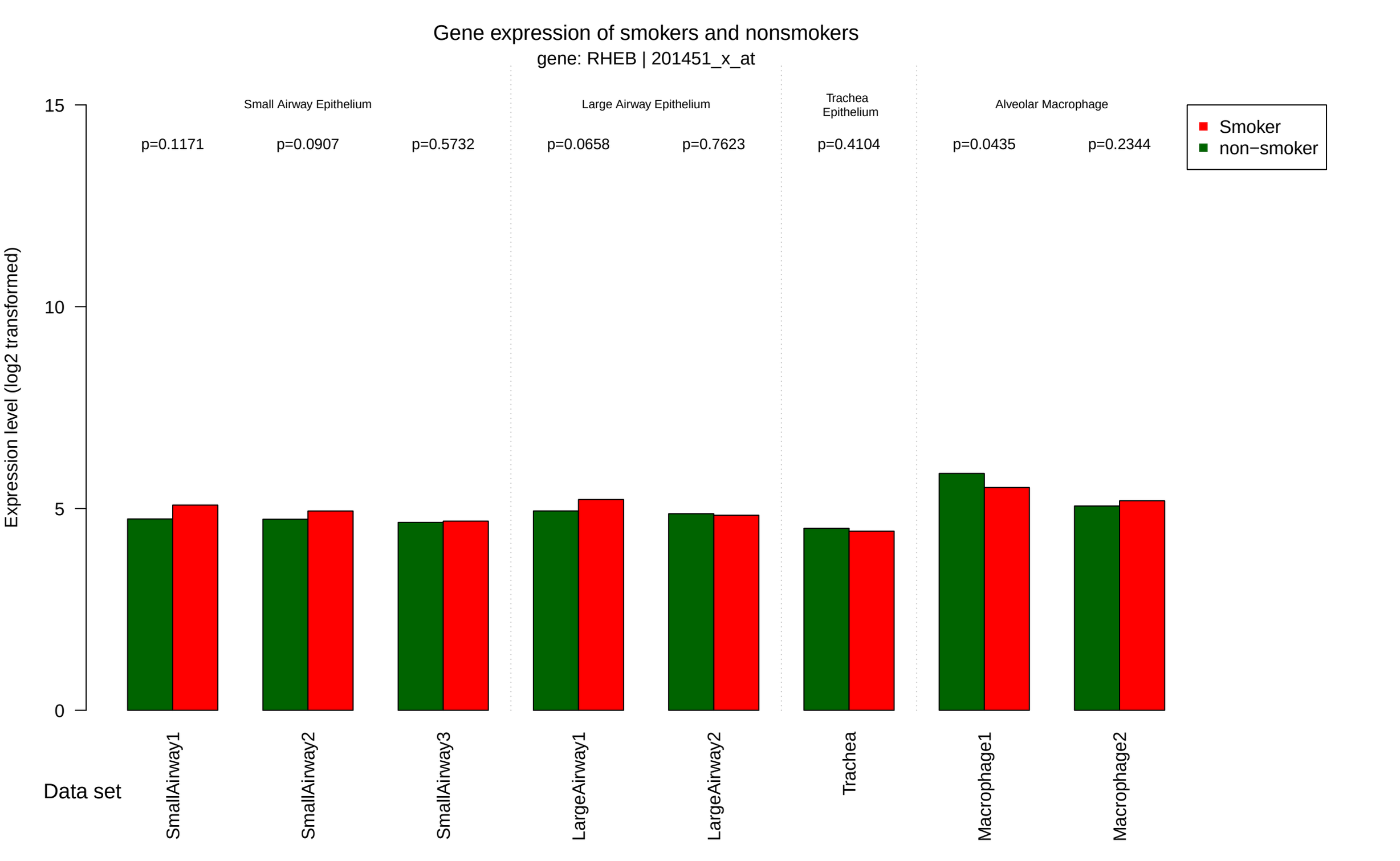
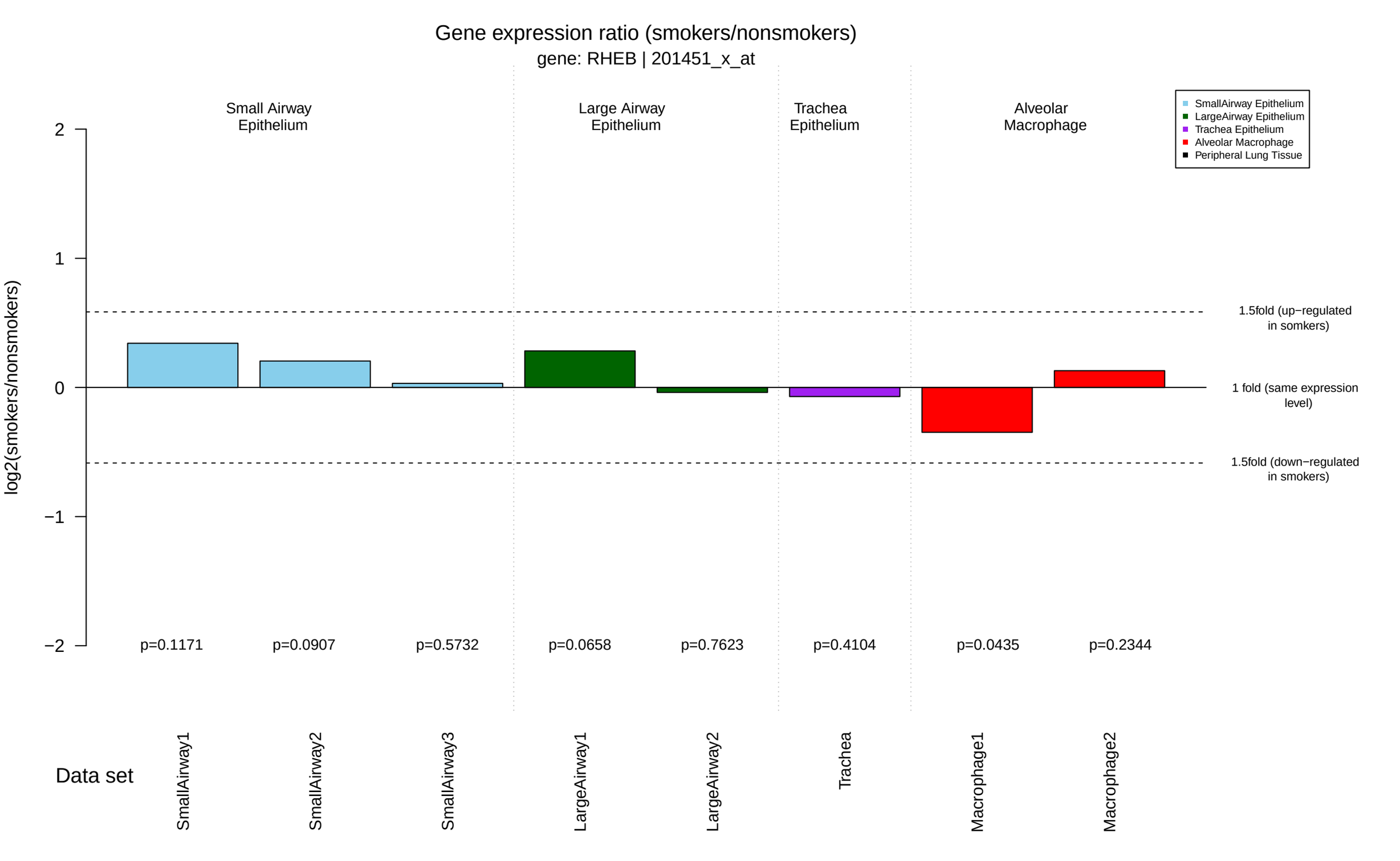
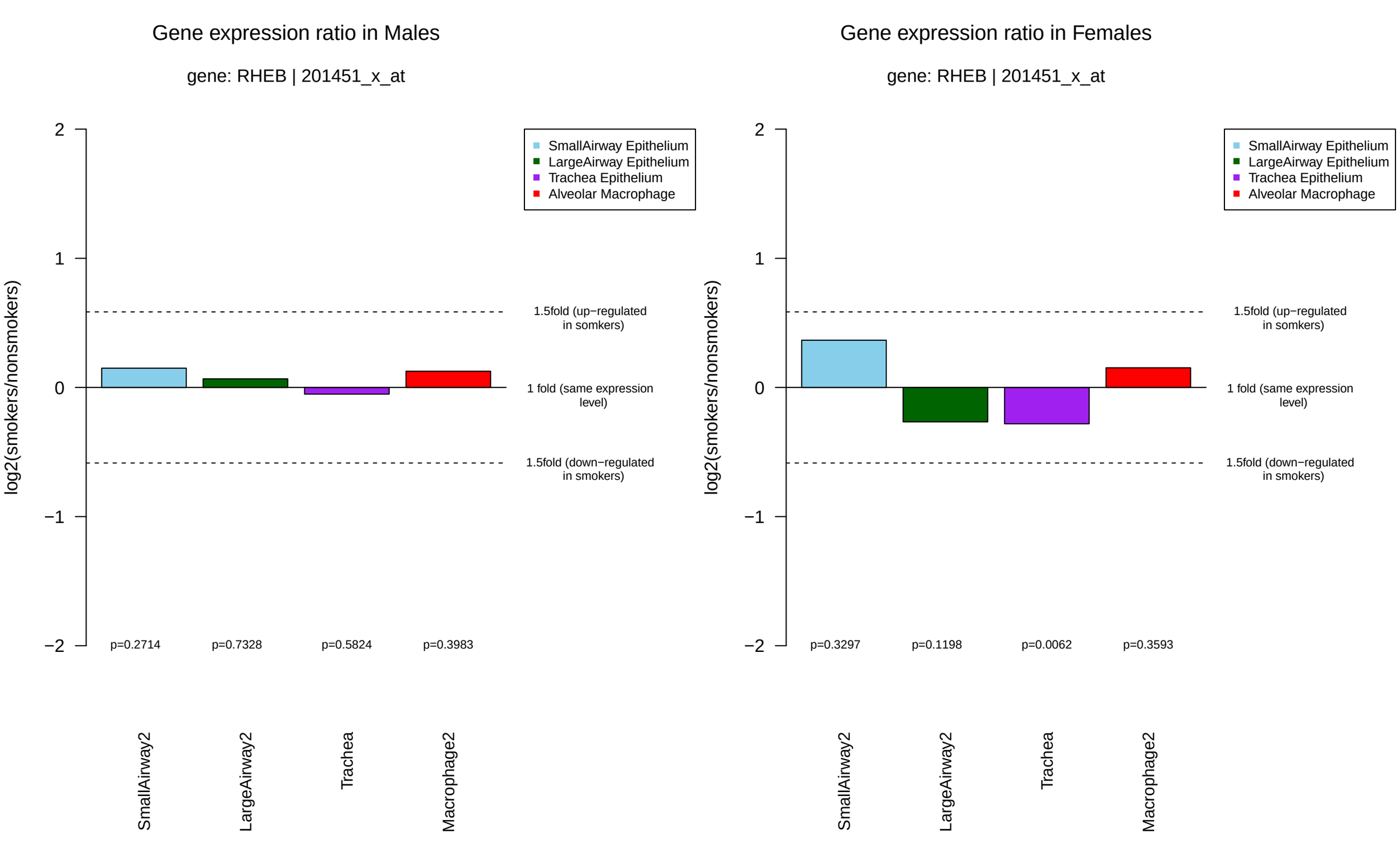
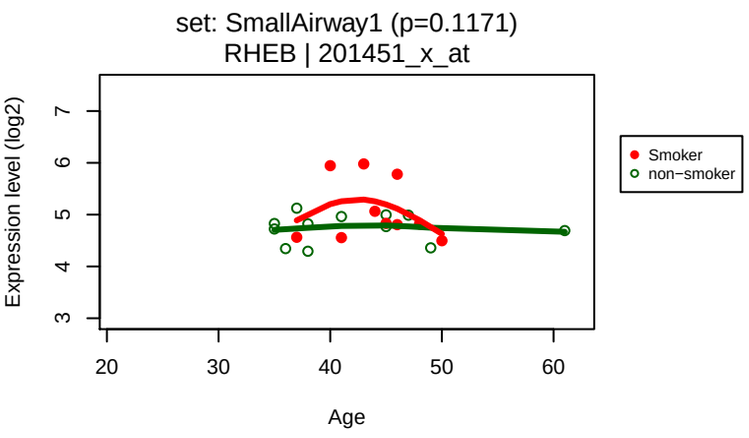
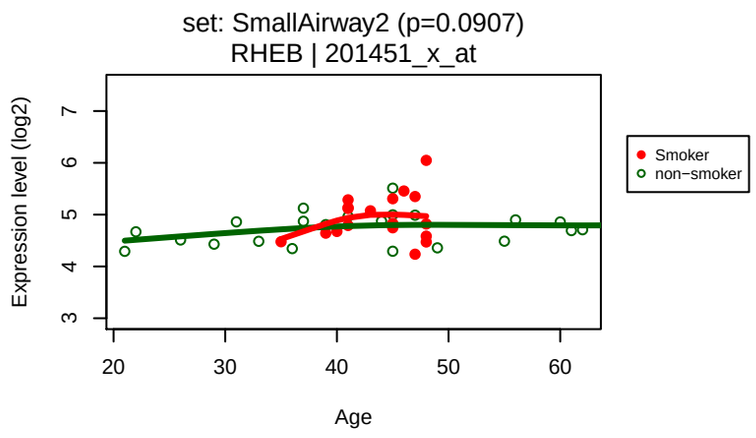
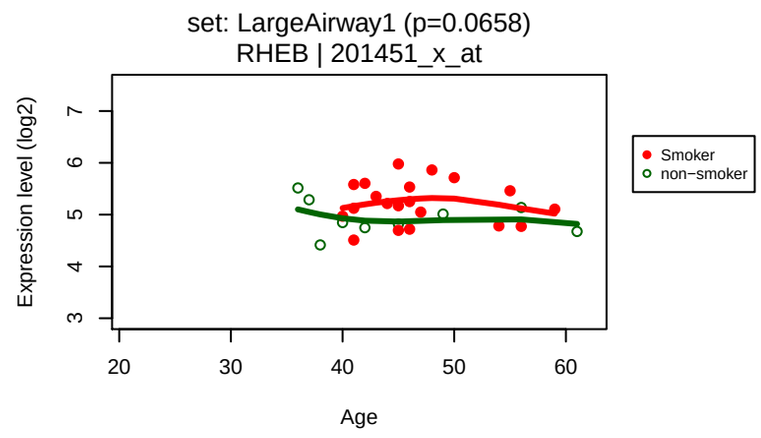

Gene expression of smokers and nonsmokers
gene: RHEB | 201451_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1171
p=0.0907
p=0.5732
p=0.0658
p=0.7623
p=0.4104
p=0.0435
p=0.2344
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

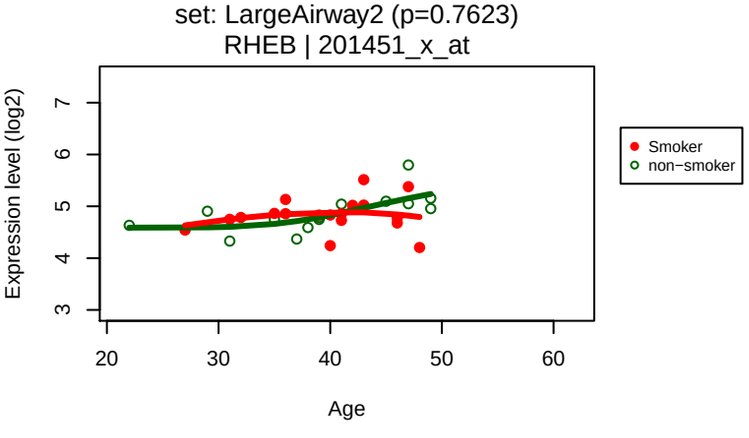

Gene expression ratio (smokers/nonsmokers)
gene: RHEB | 201451_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1171
p=0.0907
p=0.5732
p=0.0658
p=0.7623
p=0.4104
p=0.0435
p=0.2344
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

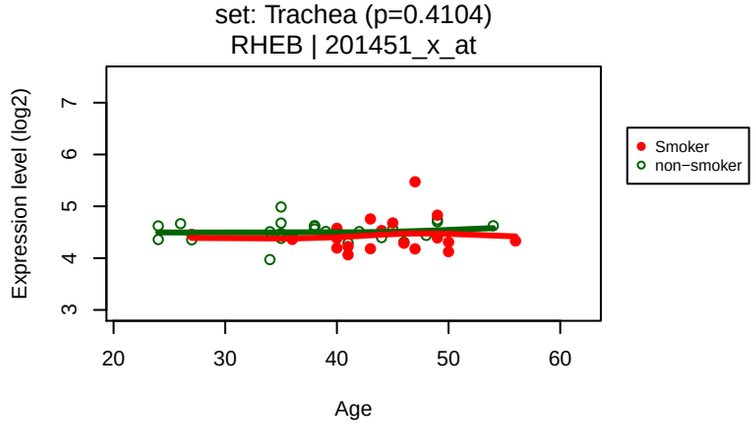

Gene expression ratio in Males
gene: RHEB | 201451_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2714
p=0.7328
p=0.5824
p=0.3983
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RHEB | 201451_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3297
p=0.1198
p=0.0062
p=0.3593
SmallAirway2
LargeAirway2
Trachea
Macrophage2

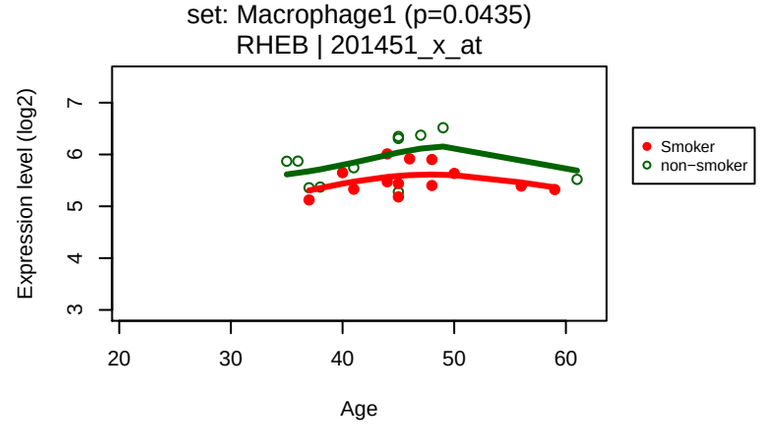

set: SmallAirway1 (p=0.1171)
RHEB | 201451_x_at
Expression level (log2)
Age
Smoker
non−smoker

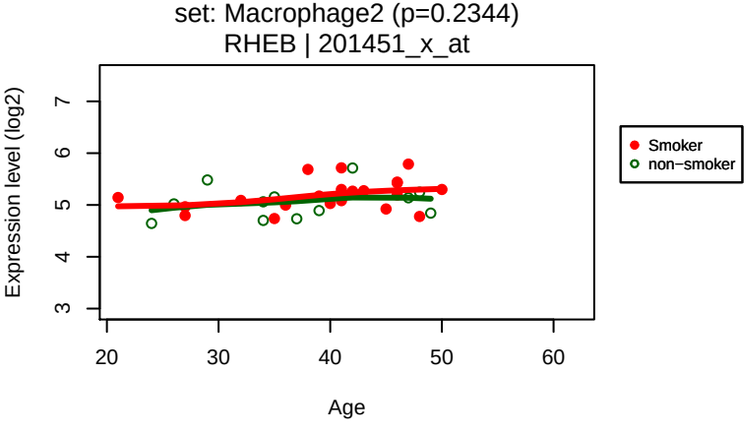

set: SmallAirway2 (p=0.0907)
RHEB | 201451_x_at
Expression level (log2)
Age
Smoker
non−smoker

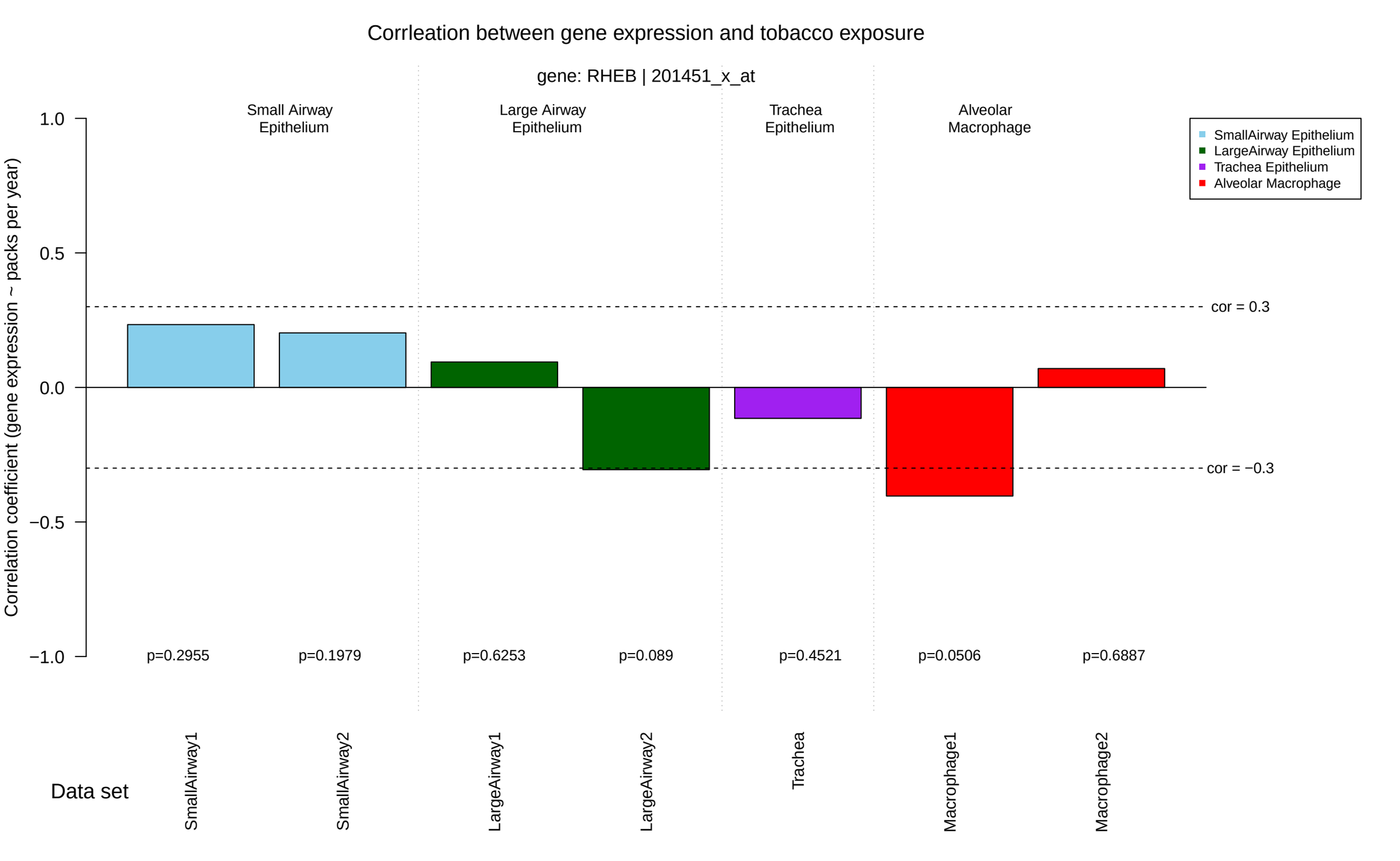

set: LargeAirway1 (p=0.0658)
RHEB | 201451_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7623)
RHEB | 201451_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4104)
RHEB | 201451_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0435)
RHEB | 201451_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2344)
RHEB | 201451_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RHEB | 201451_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2955
p=0.1979
p=0.6253
p=0.089
p=0.4521
p=0.0506
p=0.6887
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2